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VEHICLE ADAS NOTIFICATION MUTING

Abstract

A vehicle system including a vehicle including a chassis, a body assembly coupled to the chassis, a prime mover configured to generate mechanical energy to drive the vehicle, a lift assembly; and a vehicle control system including a sensor integrated into the body assembly. The vehicle control system is configured to receive sensor data from the sensor. The sensor data indicates a potential event. The vehicle control system is further configured to generate a notification associated with the potential event, receive control data indicating a state of the refuse vehicle, filter the sensor data through the control data to determine if the potential event is a false event associated with the state of the refuse vehicle or a true event associated with an external source, and mask the notification in response to a determination that the potential event is a false event.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/111,116, filed Feb. 17, 2023, which claims the benefit of and priority to U.S. Provisional Application No. 63/325,943, filed on Mar. 31, 2022, the entire contents of each of which are hereby incorporated by reference herein.

BACKGROUND

[0002] The present disclosure relates generally to vehicles. More specifically, the present disclosure relates to advanced driver assistance systems (ADAS) for vehicles.

SUMMARY

[0003] An embodiment relates to a system. The system includes a vehicle having a chassis, a body assembly coupled to the chassis, a prime mover configured to generate mechanical energy to drive the vehicle, a lift assembly, and a vehicle control system including a sensor integrated into the body assembly. The vehicle control system is configured to receive sensor data from the sensor. The sensor data can indicate a potential event. The vehicle control system is further configured to generate a notification associated with the potential event, receive control data indicating a state of the refuse vehicle, filter the sensor data through the control data to determine if the potential event is a false event associated with the state of the refuse vehicle or a true event associated with an external source, and silence the notification in response to a determination that the potential event is a false event.

[0004] An embodiment relates to a method. The method may be performed by a vehicle control system. The method includes receiving, sensor data indicating a potential event from a sensor, wherein the sensor is integrated into a body assembly coupled to a chassis of an vehicle comprising a prime mover configured to generate mechanical energy to drive the vehicle; and a lift assembly. The method includes generating a notification associated with the potential event. The method includes receiving control data indicating a state of the vehicle. The method includes filtering the sensor data through the control data to determine if the potential event is a false event associated with the state of the vehicle or a true event associated with an external source. The method includes masking the notification in response to a determination that the potential event is the false event.

[0005] An embodiment relates to a non-transitory computer-readable media. The computer-readable media includes instructions to receive sensor data indicating a potential event from a sensor; generate a notification associated with the potential event; receive control data indicating a state of a vehicle; filter the sensor data through the control data to determine if the potential event is a false event associated with the state of the vehicle or a true event associated with an external source; initiate a control action in response to the potential event, wherein the control action provides the notification to a user of the vehicle responsive to a determination that the potential event is the true event; and mask the notification in response to a determination that the potential event is the false event.

[0006] This summary is illustrative only and is not intended to be in any way limiting. Other aspects, inventive features, and advantages of the devices or processes described herein will

become apparent in the detailed description set forth herein, taken in conjunction with the accompanying figures, wherein like reference numerals refer to like elements.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. 1 is a left side view of a vehicle, according to an exemplary embodiment.

[0008] FIG. 2 is a perspective view of a chassis of the vehicle of FIG. 1.

[0009] FIG. 3 is a perspective view of the vehicle of FIG. 1 configured as a front-loading refuse vehicle, according to an exemplary embodiment.

[0010] FIG. 4 is a left side view of the front-loading refuse vehicle of FIG. 3 configured with a tag axle, according to an exemplary embodiment.

[0011] FIG. 5 is a perspective view of the vehicle of FIG. 1 configured as a side-loading refuse vehicle, according to an exemplary embodiment.

[0012] FIG. 6 is a right side view of the side-loading refuse vehicle of FIG. 5, according to an exemplary embodiment.

[0013] FIG. 7 is a top view of the side-loading refuse vehicle of FIG. 5, according to an exemplary embodiment.

[0014] FIG. 8 is a left side view of the side-loading refuse vehicle of FIG. 5 configured with a tag axle, according to an exemplary embodiment.

[0015] FIG. 9 is a perspective view of the vehicle of FIG. 1 configured as a mixer vehicle, according to an exemplary embodiment.

[0016] FIG. 10 is a perspective view of the vehicle of FIG. 1 configured as a fire fighting vehicle, according to an exemplary embodiment.

[0017] FIG. 11 is a left side view of the vehicle of FIG. 1 configured as an airport fire fighting vehicle, according to an exemplary embodiment.

[0018] FIG. 12 is a perspective view of the vehicle of FIG. 1 configured as a boom lift, according to an exemplary embodiment.

[0019] FIG. 13 is a perspective view of the vehicle of FIG. 1 configured as a scissor lift, according to an exemplary embodiment.

[0020] FIG. 14 is a top view of the vehicle of FIG. 1 equipped with an advanced driver-assistance system, including a 360-degree camera system, according to an exemplary embodiment.

[0021] FIG. 15 is a top view of the vehicle of FIG. 1 equipped with an advanced driver-assistance system, including a radar detection system, according to an exemplary embodiment.

[0022] FIG. 16 is a top view of the vehicle of FIG. 1 equipped with an advanced driver-assistance system, including a 360-degree camera system and radar detection system, according to an exemplary embodiment.

[0023] FIG. 17 is a flow diagram of a process for muting notifications regarding obstacles and/or objects in the vicinity of the vehicle of FIG. 1, according to an exemplary embodiment.

DETAILED DESCRIPTION

[0024] Before turning to the figures, which illustrate certain exemplary embodiments in detail, it should be understood that the present disclosure is not limited to the details or methodology set forth in the description or illustrated in the figures. It should also be understood that the terminology used herein is for the purpose of description only and should not be regarded as limiting.

[0025] In some ADAS systems, audio and visual notifications are provided whenever an external object or obstacle is detected near the vehicle. However, in many systems, there are both visual and audio notifications whenever a portion of the vehicle extends outward to perform a function (e.g., arm and/or grabber, extension, tailgate lift, etc.), which are a nuisance to the driver. In many cases,

this is because there is no way of distinguishing between a movement and/or presence of a portion of the vehicle and an external object or obstacle. Accordingly, there is a need to mask such notifications when the vehicle performs a function that is known to otherwise provide a notification.

[0026] According to an exemplary embodiment, a vehicle (e.g., refuse truck, mixer vehicle, fire fighting vehicle, etc.) includes a vehicle control system configured to operate as an advanced driver-assistance system (ADAS). The ADAS system includes one or more sensors positioned in and around the vocational vehicle. In one embodiment, the sensors include a three hundred sixty degree camera system and a three hundred sixty degree radar system.

Overall Vehicle

[0027] Referring to FIGS. 1 and 2, a reconfigurable vehicle (e.g., a vehicle assembly, a truck, a vehicle base, etc.) is shown as vehicle **10**, according to an exemplary embodiment. As shown, the vehicle **10** includes a frame assembly or chassis assembly, shown as chassis **20**, that supports other components of the vehicle **10**. The chassis **20** extends longitudinally along a length of the vehicle **10**, substantially parallel to a primary direction of travel of the vehicle **10**. As shown, the chassis **20** includes three sections or portions, shown as front section **22**, middle section **24**, and rear section **26**. The middle section **24** of the chassis **20** extends between the front section **22** and the rear section **26**. In some embodiments, the middle section **24** of the chassis **20** couples the front section **22** to the rear section **26**. In other embodiments, the front section **22** is coupled to the rear section **26** by another component (e.g., the body of the vehicle **10**).

[0028] As shown in FIG. 2, the front section **22** includes a pair of frame portions, frame members, or frame rails, shown as front rail portion **30** and front rail portion **32**. The rear section **26** includes a pair of frame portions, frame members, or frame rails, shown as rear rail portion **34** and rear rail portion **36**. The front rail portion **30** is laterally offset from the front rail portion **32**. Similarly, the rear rail portion **34** is laterally offset from the rear rail portion **36**. This spacing may provide frame stiffness and space for vehicle components (e.g., batteries, motors, axles, gears, etc.) between the frame rails. In some embodiments, the front rail portions **30** and **32** and the rear rail portions **34** and **36** extend longitudinally and substantially parallel to one another. The chassis **20** may include additional structural elements (e.g., cross members that extend between and couple the frame rails).

[0029] In some embodiments, the front section **22** and the rear section **26** are configured as separate, discrete subframes (e.g., a front subframe and a rear subframe). In such embodiments, the front rail portion **30**, the front rail portion **32**, the rear rail portion **34**, and the rear rail portion **36** are separate, discrete frame rails that are spaced apart from one another. In some embodiments, the front section **22** and the rear section **26** are each directly coupled to the middle section **24** such that the middle section **24** couples the front section **22** to the rear section **26**. Accordingly, the middle section **24** may include a structural housing or frame. In other embodiments, the front section **22**, the middle section **24**, and the rear section **26** are coupled to one another by another component, such as a body of the vehicle **10**.

[0030] In other embodiments, the front section **22**, the middle section **24**, and the rear section **26** are defined by a pair of frame rails that extend continuously along the entire length of the vehicle **10**. In such an embodiment, the front rail portion **30** and the rear rail portion **34** would be front and rear portions of a first frame rail, and the front rail portion **32** and the rear rail portion **36** would be front and rear portions of a second frame rail. In such embodiments, the middle section **24** would include a center portion of each frame rail.

[0031] In some embodiments, the middle section **24** acts as a storage portion that includes one or more vehicle components. The middle section **24** may include an enclosure that contains one or more vehicle components and/or a frame that supports one or more vehicle components. By way of example, the middle section **24** may contain or include one or more electrical energy storage devices (e.g., batteries, capacitors, etc.). By way of another example, the middle section **24** may include fuel tanks fuel tanks. By way of yet another example, the middle section **24** may define a

void space or storage volume that can be filled by a user.

[0032] A cabin, operator compartment, or body component, shown as cab **40**, is coupled to a front end portion of the chassis **20** (e.g., the front section **22** of the chassis **20**). Together, the chassis **20** and the cab **40** define a front end of the vehicle **10**. The cab **40** extends above the chassis **20**. The cab **40** includes an enclosure or main body that defines an interior volume, shown as cab interior **42**, that is sized to contain one or more operators. The cab **40** also includes one or more doors **44** that facilitate selective access to the cab interior **42** from outside of the vehicle **10**. The cab interior **42** contains one or more components that facilitate operation of the vehicle **10** by the operator. By way of example, the cab interior **42** may contain components that facilitate operator comfort (e.g., seats, seatbelts, etc.), user interface components that receive inputs from the operators (e.g., steering wheels, pedals, touch screens, switches, buttons, levers, etc.), and/or user interface components that provide information to the operators (e.g., lights, gauges, speakers, etc.). The user interface components within the cab **40** may facilitate operator control over the drive components of the vehicle **10** and/or over any implements of the vehicle **10**.

[0033] The vehicle **10** further includes a series of axle assemblies, shown as front axle **50** and rear axles **52**. As shown, the vehicle **10** includes one front axle **50** coupled to the front section **22** of the chassis **20** and two rear axles **52** each coupled to the rear section **26** of the chassis **20**. In other embodiments, the vehicle **10** includes more or fewer axles. By way of example, the vehicle **10** may include a tag axle that may be raised or lowered to accommodate variations in weight being carried by the vehicle **10**. The front axle **50** and the rear axles **52** each include a series of tractive elements (e.g., wheels, treads, etc.), shown as wheel and tire assemblies **54**. The wheel and tire assemblies **54** are configured to engage a support surface (e.g., roads, the ground, etc.) to support and propel the vehicle **10**. The front axle **50** and the rear axles may include steering components (e.g., steering arms, steering actuators, etc.), suspension components (e.g., gas springs, dampeners, air springs, etc.), power transmission or drive components (e.g., differentials, drive shafts, etc.), braking components (e.g., brake actuators, brake pads, brake discs, brake drums, etc.), and/or other components that facilitate propulsion or support of the vehicle.

[0034] In some embodiments, the vehicle **10** is configured as an electric vehicle that is propelled by an electric powertrain system. Referring to FIG. **1**, the vehicle **10** includes one or more electrical energy storage devices (e.g., batteries, capacitors, etc.), shown as batteries **60**. As shown, the batteries **60** are positioned within the middle section **24** of the chassis **20**. In other embodiments, the batteries **60** are otherwise positioned throughout the vehicle **10**. The vehicle **10** further includes one or more electromagnetic devices or prime movers (e.g., motor/generators), shown as drive motors **62**. The drive motors **62** are electrically coupled to the batteries **60**. The drive motors **62** may be configured to receive electrical energy from the batteries **60** and provide rotational mechanical energy to the wheel and tire assemblies **54** to propel the vehicle **10**. The drive motors **62** may be configured to receive rotational mechanical energy from the wheel and tire assemblies **54** and provide electrical energy to the batteries **60**, providing a braking force to slow the vehicle **10**.

[0035] The batteries **60** may include one or more rechargeable batteries (e.g., lithium-ion batteries, nickel-metal hydride batteries, lithium-ion polymer batteries, lead-acid batteries, nickel-cadmium batteries, etc.). The batteries **60** may be charged by one or more sources of electrical energy onboard the vehicle **10** (e.g., solar panels, etc.) or separate from the vehicle **10** (e.g., connections to an electrical power grid, a wireless charging system, etc.). As shown, the drive motors **62** are positioned within the rear axles **52** (e.g., as part of a combined axle and motor assembly). In other embodiments, the drive motors **62** are otherwise positioned within the vehicle **10**.

[0036] In other embodiments, the vehicle **10** is configured as a hybrid vehicle that is propelled by a hybrid powertrain system (e.g., a diesel/electric hybrid, gasoline/electric hybrid, natural gas/electric hybrid, etc.). According to an exemplary embodiment, the hybrid powertrain system may include a primary driver (e.g., an engine, a motor, etc.), an energy generation device (e.g., a generator, etc.),

and/or an energy storage device (e.g., a battery, capacitors, ultra-capacitors, etc.) electrically coupled to the energy generation device. The primary driver may combust fuel (e.g., gasoline, diesel, etc.) to provide mechanical energy, which a transmission may receive and provide to the axle front axle **50** and/or the rear axles **52** to propel the vehicle **10**. Additionally or alternatively, the primary driver may provide mechanical energy to the generator, which converts the mechanical energy into electrical energy. The electrical energy may be stored in the energy storage device (e.g., the batteries **60**) in order to later be provided to a motive driver.

[0037] In yet other embodiments, the chassis **20** may further be configured to support non-hybrid powertrains. For example, the powertrain system may include a primary driver that is a compression-ignition internal combustion engine that utilizes diesel fuel.

[0038] Referring to FIG. **1**, the vehicle **10** includes a rear assembly, module, implement, body, or cargo area, shown as application kit **80**. The application kit **80** may include one or more implements, vehicle bodies, and/or other components. Although the application kit **80** is shown positioned behind the cab **40**, in other embodiments the application kit **80** extends forward of the cab **40**. The vehicle **10** may be outfitted with a variety of different application kits **80** to configure the vehicle **10** for use in different applications. Accordingly, a common vehicle **10** can be configured for a variety of different uses simply by selecting an appropriate application kit **80**. By way of example, the vehicle **10** may be configured as a refuse vehicle, a concrete mixer, a fire fighting vehicle, an airport fire fighting vehicle, a lift device (e.g., a boom lift, a scissor lift, a telehandler, a vertical lift, etc.), a crane, a tow truck, a military vehicle, a delivery vehicle, a mail vehicle, a boom truck, a plow truck, a farming machine or vehicle, a construction machine or vehicle, a coach bus, a school bus, a semi-truck, a passenger or work vehicle (e.g., a sedan, a SUV, a truck, a van, etc.), and/or still another vehicle. FIGS. **3-13** illustrate various examples of how the vehicle **10** may be configured for specific applications. Although only a certain set of vehicle configurations is shown, it should be understood that the vehicle **10** may be configured for use in other applications that are not shown. Further, references to embodiments employing a particular type of vehicle **10** may employ other types of vehicles. For example, references to an ADAS of a refuse vehicle **10** detecting a portion of the vehicle is not intended to be limiting; other embodiments may employ a concrete truck, firefighting vehicle, sedan, or the like.

[0039] The application kit **80** may include various actuators to facilitate certain functions of the vehicle **10**. By way of example, the application kit **80** may include hydraulic actuators (e.g., hydraulic cylinders, hydraulic motors, etc.), pneumatic actuators (e.g., pneumatic cylinders, pneumatic motors, etc.), and/or electrical actuators (e.g., electric motors, electric linear actuators, etc.). The application kit **80** may include components that facilitate operation of and/or control of these actuators. By way of example, the application kit **80** may include hydraulic or pneumatic components that form a hydraulic or pneumatic circuit (e.g., conduits, valves, pumps, compressors, gauges, reservoirs, accumulators, etc.). By way of another example, the application kit **80** may include electrical components (e.g., batteries, capacitors, voltage regulators, motor controllers, etc.). The actuators may be powered by components of the vehicle **10**. By way of example, the actuators may be powered by the batteries **60**, the drive motors **62**, or the primary driver (e.g., through a power take off).

[0040] The vehicle **10** generally extends longitudinally from a front side **86** to a rear side **88**. The front side **86** is defined by the cab **40** and/or the chassis. The rear side **88** is defined by the application kit **80** and/or the chassis **20**. The primary, forward direction of travel of the vehicle **10** is longitudinal, with the front side **86** being arranged forward of the rear side **88**.

A. Front-Loading Refuse Vehicle

[0041] Referring now to FIGS. **3** and **4**, the vehicle **10** is configured as a refuse vehicle **100** (e.g., a refuse truck, a garbage truck, a waste collection truck, a sanitation truck, a recycling truck, etc.). Specifically, the refuse vehicle **100** is a front-loading refuse vehicle. In other embodiments, the refuse vehicle **100** is configured as a rear-loading refuse vehicle or a front-loading refuse vehicle.

The refuse vehicle **100** may be configured to transport refuse from various waste receptacles (e.g., refuse containers) within a municipality to a storage and/or processing facility (e.g., a landfill, an incineration facility, a recycling facility, etc.).

[0042] FIG. **4** illustrates the refuse vehicle **100** of FIG. **3** configured with a liftable axle, shown as tag axle **90**, including a pair of wheel and tire assemblies **54**. As shown, the tag axle **90** is positioned rearward of the rear axles **52**. The tag axle **90** can be selectively raised and lowered (e.g., by a hydraulic actuator) to selectively engage the wheel and tire assemblies **54** of the tag axle **90** with the ground. The tag axle **90** may be raised to reduce rolling resistance experienced by the refuse vehicle **100**. The tag axle **90** may be lowered to distribute the loaded weight of the vehicle **100** across a greater number of a wheel and tire assemblies **54** (e.g., when the refuse vehicle **100** is loaded with refuse).

[0043] As shown in FIGS. **3** and **4**, the application kit **80** of the refuse vehicle **100** includes a series of panels that form a rear body or container, shown as refuse compartment **130**. The refuse compartment **130** may facilitate transporting refuse from various waste receptacles within a municipality to a storage and/or a processing facility (e.g., a landfill, an incineration facility, a recycling facility, etc.). By way of example, loose refuse may be placed into the refuse compartment **130** where it may be compacted (e.g., by a packer system within the refuse compartment **130**). The refuse compartment **130** may also provide temporary storage for refuse during transport to a waste disposal site and/or a recycling facility. In some embodiments, the refuse compartment **130** may define a hopper volume **132** and storage volume **134**. In this regard, refuse may be initially loaded into the hopper volume **132** and later compacted into the storage volume **134**. As shown, the hopper volume **132** is positioned between the storage volume **134** and the cab **40** (e.g., refuse is loaded into a portion of the refuse compartment **130** behind the cab **40** and stored in a portion further toward the rear of the refuse compartment **130**). In other embodiments, the storage volume may be positioned between the hopper volume and the cab **40** (e.g., in a rear-loading refuse truck, etc.). The application kit **80** of the refuse vehicle **100** further includes a pivotable rear portion, shown as tailgate **136**, that is pivotally coupled to the refuse compartment **130**. The tailgate **136** may be selectively repositionable between a closed position and an open position by an actuator (e.g., a hydraulic cylinder, an electric linear actuator, etc.), shown as tailgate actuator **138** (e.g., to facilitate emptying the storage volume).

[0044] As shown in FIGS. **3** and **4**, the refuse vehicle **100** also includes an implement, shown as lift assembly **140**, which is a front-loading lift assembly. According to an exemplary embodiment, the lift assembly **140** includes a pair of lift arms **142** and a pair of actuators (e.g., hydraulic cylinders, electric linear actuators, etc.), shown as lift arm actuators **144**. The lift arms **142** may be rotatably coupled to the chassis **20** and/or the refuse compartment **130** on each side of the refuse vehicle **100** (e.g., through a pivot, a lug, a shaft, etc.), such that the lift assembly **140** may extend forward relative to the cab **40** (e.g., a front-loading refuse truck, etc.). In other embodiments, the lift assembly **140** may extend rearward relative to the application kit **80** (e.g., a rear-loading refuse truck). As shown in FIGS. **3** and **4**, in an exemplary embodiment the lift arm actuators **144** may be positioned such that extension and retraction of the lift arm actuators **144** rotates the lift arms **142** about an axis extending through the pivot. In this regard, the lift arms **142** may be rotated by the lift arm actuators **144** to lift a refuse container over the cab **40**. The lift assembly **140** further includes a pair of interface members, shown as lift forks **146**, each pivotally coupled to a distal end of one of the lift arms **142**. The lift forks **146** may be configured to engage a refuse container (e.g., a dumpster) to selectively couple the refuse container to the lift arms **142**. By way of example, each of the lift forks **146** may be received within a corresponding pocket defined by the refuse container. A pair of actuators (e.g., hydraulic cylinders, electric linear actuators, etc.), shown as articulation actuators **148**, are each coupled to one of the lift arms **142** and one of the lift forks **146**. The articulation actuators **148** may be positioned to rotate the lift forks **146** relative to the lift arms **142** about a horizontal axis. Accordingly, the articulation actuators **148** may assist in tipping refuse out

of the refuse container and into the refuse compartment **130**. The lift arm actuators **144** may then rotate the lift arms **142** to return the empty refuse container to the ground.

B. Side-Loading Refuse Vehicle

[0045] Referring now to FIGS. 5-8, an alternative configuration of the refuse vehicle **100** is shown according to an exemplary embodiment. Specifically, the refuse vehicle **100** of FIGS. 5-8 is configured as a side-loading refuse vehicle. The refuse vehicle **100** of FIGS. 5-8 may be substantially similar to the front-loading refuse vehicle **100** of FIGS. 3 and 4 except as otherwise specified herein. As shown, the refuse vehicle **100** of FIGS. 5-7 is configured with a tag axle **90** in FIG. 8.

[0046] Referring still to FIGS. 5-8, the refuse vehicle **100** omits the lift assembly **140** and instead includes a side-loading lift assembly, shown as lift assembly **160**, that extends laterally outward from a side of the refuse vehicle **100**. The lift assembly **160** includes an interface assembly, shown as grabber assembly **162**, that is configured to engage a refuse container (e.g., a residential garbage can) to selectively couple the refuse container to the lift assembly **160**. The grabber assembly **162** includes a main portion, shown as main body **164**, and a pair of fingers or interface members, shown as grabber fingers **166**. The grabber fingers **166** are pivotally coupled to the main body **164** such that the grabber fingers **166** are each rotatable about a vertical axis. A pair of actuators (e.g., hydraulic motors, electric motors, etc.), shown as finger actuators **168**, are configured to control movement of the grabber fingers **166** relative to the main body **164**.

[0047] The grabber assembly **162** is movably coupled to a guide, shown as track **170**, that extends vertically along a side of the refuse vehicle **100**. Specifically, the main body **164** is slidably coupled to the track **170** such that the main body **164** is repositionable along a length of the track **170**. An actuator (e.g., a hydraulic motor, an electric motor, etc.), shown as lift actuator **172**, is configured to control movement of the grabber assembly **162** along the length of the track **170**. In some embodiments, a bottom end portion of the track **170** is straight and substantially vertical such that the grabber assembly **162** raises or lowers a refuse container when moving along the bottom end portion of the track **170**. In some embodiments, a top end portion of the track **170** is curved such that the grabber assembly **162** inverts a refuse container to dump refuse into the hopper volume **132** when moving along the top end portion of the track **170**.

[0048] The lift assembly **160** further includes an actuator (e.g., a hydraulic cylinder, an electric linear actuator, etc.), shown as track actuator **174**, that is configured to control lateral movement of the grabber assembly **162**. By way of example, the track actuator **174** may be coupled to the chassis **20** and the track **170** such that the track actuator **174** moves the track **170** and the grabber assembly **162** laterally relative to the chassis **20**. The track actuator **174** may facilitate repositioning the grabber assembly **162** to pick up and replace refuse containers that are spaced laterally outward from the refuse vehicle **100**.

C. Concrete Mixer Truck

[0049] Referring now to FIG. 9, the vehicle **10** is configured as a mixer truck (e.g., a concrete mixer truck, a mixer vehicle, etc.), shown as mixer truck **200**. Specifically, the mixer truck **200** is shown as a rear-discharge concrete mixer truck. In other embodiments, the mixer truck **200** is a front-discharge concrete mixer truck.

[0050] As shown in FIG. 9, the application kit **80** includes a mixing drum assembly (e.g., a concrete mixing drum), shown as drum assembly **230**. The drum assembly **230** may include a mixing drum **232**, a drum drive system **234** (e.g., a rotational actuator or motor, such as an electric motor or hydraulic motor), an inlet portion, shown as hopper **236**, and an outlet portion, shown as chute **238**. The mixing drum **232** may be coupled to the chassis **20** and may be disposed behind the cab **40** (e.g., at the rear and/or middle of the chassis **20**). In an exemplary embodiment, the drum drive system **234** is coupled to the chassis **20** and configured to selectively rotate the mixing drum **232** about a central, longitudinal axis. According to an exemplary embodiment, the central, longitudinal axis of the mixing drum **232** may be elevated from the chassis **20** (e.g., from a

horizontal plan extending along the chassis **20**) at an angle in the range of five degrees to twenty degrees. In other embodiments, the central, longitudinal axis may be elevated by less than five degrees (e.g., four degrees, etc.). In yet another embodiment, the mixer truck **200** may include an actuator positioned to facilitate adjusting the central, longitudinal axis to a desired or target angle (e.g., manually in response to an operator input/command, automatically according to a control system, etc.).

[0051] The mixing drum **232** may be configured to receive a mixture, such as a concrete mixture (e.g., cementitious material, aggregate, sand, etc.), through the hopper **236**. In some embodiments, the mixer truck **200** includes an injection system (e.g., a series of nozzles, hoses, and/or valves) including an injection valve that selectively fluidly couples a supply of fluid to the inner volume of the mixing drum **232**. By way of example, the injection system may be used to inject water and/or chemicals (e.g., air entrainers, water reducers, set retarders, set accelerators, superplasticizers, corrosion inhibitors, coloring, calcium chloride, minerals, and/or other concrete additives, etc.) into the mixing drum **232**. The injection valve may facilitate injecting water and/or chemicals from a fluid reservoir (e.g., a water tank, etc.) into the mixing drum **232**, while preventing the mixture in the mixing drum **232** from exiting the mixing drum **232** through the injection system. In some embodiments, one or more mixing elements (e.g., fins, etc.) may be positioned in the interior of the mixing drum **232**, and may be configured to agitate the contents of the mixture when the mixing drum **232** is rotated in a first direction (e.g., counterclockwise, clockwise, etc.), and drive the mixture out through the chute **238** when the mixing drum **232** is rotated in a second direction (e.g., clockwise, counterclockwise, etc.). In some embodiments, the chute **238** may also include an actuator positioned such that the chute **238** may be selectively pivotable to position the chute **238** (e.g., vertically, laterally, etc.), for example at an angle at which the mixture is expelled from the mixing drum **232**.

D. Fire Truck

[0052] Referring now to FIG. **10**, the vehicle **10** is configured as a fire fighting vehicle, fire truck, or fire apparatus (e.g., a turntable ladder truck, a pumper truck, a quint, etc.), shown as fire fighting vehicle **250**. In the embodiment shown in FIG. **10**, the fire fighting vehicle **250** is configured as a rear-mount aerial ladder truck. In other embodiments, the fire fighting vehicle **250** is configured as a mid-mount aerial ladder truck, a quint fire truck (e.g., including an on-board water storage, a hose storage, a water pump, etc.), a tiller fire truck, a pumper truck (e.g., without an aerial ladder), or another type of response vehicle. By way of example, the vehicle **10** may be configured as a police vehicle, an ambulance, a tow truck, or still other vehicles used for responding to a scene (e.g., an accident, a fire, an incident, etc.).

[0053] As shown in FIG. **10**, in the fire fighting vehicle **250**, the application kit **80** is positioned mainly rearward from the cab **40**. The application kit **80** includes deployable stabilizers (e.g., outriggers, downriggers, etc.), shown as outriggers **252**, that are coupled to the chassis **20**. The outriggers **252** may be configured to selectively extend from each lateral side and/or the rear of the fire fighting vehicle **250** and engage a support surface (e.g., the ground) in order to provide increased stability while the fire fighting vehicle **250** is stationary. The fire fighting vehicle **250** further includes an extendable or telescoping ladder assembly, shown as ladder assembly **254**. The increased stability provided by the outriggers **252** is desirable when the ladder assembly **254** is in use (e.g., extended from the fire fighting vehicle **250**) to prevent tipping. In some embodiments, the application kit **80** further includes various storage compartments (e.g., cabinets, lockers, etc.) that may be selectively opened and/or accessed for storage and/or component inspection, maintenance, and/or replacement.

[0054] As shown in FIG. **10**, the ladder assembly **254** includes a series of ladder sections **260** that are slidably coupled with one another such that the ladder sections **260** may extend and/or retract (e.g., telescope) relative to one another to selectively vary a length of the ladder assembly **254**. A base platform, shown as turntable **262**, is rotatably coupled to the chassis **20** and to a proximal end

of a base ladder section **260** (i.e., the most proximal of the ladder sections **260**). The turntable **262** may be configured to rotate about a vertical axis relative to the chassis **20** to rotate the ladder sections **260** about the vertical axis (e.g., up to 360 degrees, etc.). The ladder sections **260** may rotate relative to the turntable **262** about a substantially horizontal axis to selectively raise and lower the ladder sections **260** relative to the chassis **20**. As shown, a water turret or implement, shown as monitor **264**, is coupled to a distal end of a fly ladder section **260** (i.e., the most distal of the ladder sections **260**). The monitor **264** may be configured to expel water and/or a fire suppressing agent (e.g., foam, etc.) from a water storage tank and/or an agent tank onboard the fire fighting vehicle **250**, and/or from an external source (e.g., a fire hydrant, a separate water/pumper truck, etc.). In some embodiments, the ladder assembly **254** further includes an aerial platform coupled to the distal end of the fly ladder section **260** and configured to support one or more operators.

E. ARFF Truck

[0055] Referring now to FIG. **11**, the vehicle **10** is configured as a fire fighting vehicle, shown as airport rescue and fire fighting (ARFF) truck **300**. As shown in FIG. **11**, the application kit **80** is positioned primarily rearward of the cab **40**. As shown, the application kit **80** includes a series of storage compartments or cabinets, shown as compartments **302**, that are coupled to the chassis **20**. The compartments **302** may store various equipment or components of the ARFF truck **300**.

[0056] The application kit **80** includes a pump system **304** (e.g., an ultra-high-pressure pump system, etc.) positioned within one of the compartments **302** near the center of the ARFF truck **300**. The application kit **80** further includes a water tank **310**, an agent tank **312**, and an implement or water turret, shown as monitor **314**. The pump system **304** may include a high pressure pump and/or a low pressure pump, which may be fluidly coupled to the water tank **310** and/or the agent tank **312**. The pump system **304** may pump water and/or fire suppressing agent from the water tank **310** and the agent tank **312**, respectively, to the monitor **314**. The monitor **314** may be selectively reoriented by an operator to adjust a direction of a stream of water and/or agent. As shown in FIG. **11**, the monitor **314** is coupled to a front end of the cab **40**.

F. Boom Lift

[0057] Referring now to FIG. **12**, the vehicle **10** is configured as a lift device, shown as boom lift **350**. The boom lift **350** may be configured to support and elevate one or more operators. In other embodiments, the vehicle **10** is configured as another type of lift device that is configured to lift operators and/or material, such as a skid-loader, a telehandler, a scissor lift, a fork lift, a vertical lift, and/or any other type of lift device or machine.

[0058] As shown in FIG. **12**, the application kit **80** includes a base assembly, shown as turntable **352**, that is rotatably coupled to the chassis **20**. The turntable **352** may be configured to selectively rotate relative to the chassis **20** about a substantially vertical axis. In some embodiments, the turntable **352** includes a counterweight (e.g., the batteries) positioned near the rear of the turntable **352**. The turntable **352** is rotatably coupled to a lift assembly, shown as boom assembly **354**. The boom assembly **354** includes a first section or telescoping boom section, shown as lower boom **360**. The lower boom **360** includes a series of nested boom sections that extend and retract (e.g., telescope) relative to one another to vary a length of the boom assembly **354**. The boom assembly **354** further includes a second boom section or four bar linkage, shown as upper boom **362**. The upper boom **362** may include structural members that rotate relative to one another to raise and lower a distal end of the boom assembly **354**. In other embodiments, the boom assembly **354** includes more or fewer boom sections (e.g., one, three, five, etc.) and/or a different arrangement of boom sections.

[0059] As shown in FIG. **12**, the boom assembly **354** includes a first actuator, shown as lower lift cylinder **364**. The lower boom **360** is pivotally coupled (e.g., pinned, etc.) to the turntable **352** at a joint or lower boom pivot point. The lower lift cylinder **364** (e.g., a pneumatic cylinder, an electric linear actuator, a hydraulic cylinder, etc.) is coupled to the turntable **352** at a first end and coupled

to the lower boom **360** at a second end. The lower lift cylinder **364** may be configured to raise and lower the lower boom **360** relative to the turntable **352** about the lower boom pivot point. [0060] The boom assembly **354** further includes a second actuator, shown as upper lift cylinder **366**. The upper boom **362** is pivotally coupled (e.g., pinned) to the upper end of the lower boom **360** at a joint or upper boom pivot point. The upper lift cylinder **366** (e.g., a pneumatic cylinder, an electric linear actuator, a hydraulic cylinder, etc.) is coupled to the upper boom **362**. The upper lift cylinder **366** may be configured to extend and retract to actuate (e.g., lift, rotate, elevate, etc.) the upper boom **362**, thereby raising and lowering a distal end of the upper boom **362**.

[0061] Referring still to FIG. **12**, the application kit **80** further includes an operator platform, shown as platform assembly **370**, coupled to the distal end of the upper boom **362** by an extension arm, shown as jib arm **372**. The jib arm **372** may be configured to pivot the platform assembly **370** about a lateral axis (e.g., to move the platform assembly **370** up and down, etc.) and/or about a vertical axis (e.g., to move the platform assembly **370** left and right, etc.).

[0062] The platform assembly **370** provides a platform configured to support one or more operators or users. In some embodiments, the platform assembly **370** may include accessories or tools configured for use by the operators. For example, the platform assembly **370** may include pneumatic tools (e.g., an impact wrench, airbrush, nail gun, ratchet, etc.), plasma cutters, welders, spotlights, etc. In some embodiments, the platform assembly **370** includes a control panel (e.g., a user interface, a removable or detachable control panel, etc.) configured to control operation of the boom lift **350** (e.g., the turntable **352**, the boom assembly **354**, etc.) from the platform assembly **370** or remotely. In other embodiments, the platform assembly **370** is omitted, and the boom lift **350** includes an accessory and/or tool (e.g., forklift forks, etc.) coupled to the distal end of the boom assembly **354**.

G. Scissor Lift

[0063] Referring now to FIG. **13**, the vehicle **10** is configured as a lift device, shown as scissor lift **400**. As shown in FIG. **13**, the application kit **80** includes a body, shown as lift base **402**, coupled to the chassis **20**. The lift base **402** is coupled to a scissor assembly, shown as cab **404**, such that the lift base **402** supports the cab **404**. The cab **404** is configured to extend and retract, raising and lowering between a raised position and a lowered position relative to the lift base **402**.

[0064] As shown in FIG. **13**, the lift base **402** includes a series of actuators, stabilizers, downriggers, or outriggers, shown as lift assembly **140**. The lift assembly **140** may extend and retract vertically between a stored position and a deployed position. In the stored position, the lift assembly **140** may be raised, such that the lift assembly **140** do not contact the ground. Conversely, in the deployed position, the lift assembly **140** may engage the ground to lift the lift base **402**. The length of each of the lift assembly **140** in their respective deployed positions may be varied in order to adjust the pitch (e.g., rotational position about a lateral axis) and the roll (e.g., rotational position about a longitudinal axis) of the lift base **402** and/or the chassis **20**. Accordingly, the lengths of the lift assembly **140** in their respective deployed positions may be adjusted to level the lift base **402** with respect to the direction of gravity (e.g., on uneven, sloped, pitted, etc. terrain). The lift assembly **140** may lift the wheel and tire assemblies **54** off of the ground to prevent movement of the scissor lift **400** during operation. In other embodiments, the lift assembly **140** are omitted.

[0065] The cab **404** may include a series of subassemblies, shown as scissor layers **420**, each including a pair of inner members and a pair of outer members pivotally coupled to one another. The scissor layers **420** may be stacked atop one another in order to form the cab **404**, such that movement of one scissor layer **420** causes a similar movement in all of the other scissor layers **420**. The scissor layers **420** extend between and couple the lift base **402** and an operator platform (e.g., the platform assembly **430**). In some embodiments, scissor layers **420** may be added to, or removed from, the cab **404** in order to increase, or decrease, the fully extended height of the cab **404**.

[0066] Referring still to FIG. **13**, the cab **404** may also include one or more lift actuators **424** (e.g., hydraulic cylinders, pneumatic cylinders, electric linear actuators such as motor-driven leadscrews,

etc.) configured to extend and retract the cab **404**. The lift actuators **424** may be pivotally coupled to inner members of various scissor layers **420**, or otherwise arranged within the cab **404**. [0067] A distal or upper end of the cab **404** is coupled to an operator platform, shown as platform assembly **430**. The platform assembly **430** may perform similar functions to the platform assembly **370**, such as supporting one or more operators, accessories, and/or tools. The platform assembly **430** may include a control panel to control operation of the scissor lift **400**. The lift actuators **424** may be configured to actuate the cab **404** to selectively reposition the platform assembly **430** between a lowered position (e.g., where the platform assembly **430** is proximate to the lift base **402**) and a raised position (e.g., where the platform assembly **430** is at an elevated height relative to the lift base **402**). Specifically, in some embodiments, extension of the lift actuators **424** moves the platform assembly **430** upward (e.g., extending the cab **404**), and retraction of the lift actuators **424** moves the platform assembly **430** downward (e.g., retracting the cab **404**). In other embodiments, extension of the lift actuators **424** retracts the cab **404**, and retraction of the lift actuators **424** extends the cab **404**.

External Object ADAS Notifications

[0068] According to an exemplary embodiment shown in FIG. **14**, an ADAS includes a three hundred and sixty degree (360°) camera system, shown as **360** system **500**, integrated into the refuse vehicle **10**. The 360° system **500** includes sensors, shown as cameras **510**. In some embodiments cameras **510** may be image sensors configured to capture live video and image data and provide the sensor data to ADAS. Cameras **510** have a field of view, shown as camera FOV **520**. In some embodiments, camera FOV **520** can be between 100-200 degrees. For example, cameras **510** can have a 160-degree fOV or a 190-degree fOV. The camera FOV **520** of cameras **510** may overlap on the corners to aid in stitching the various feeds together to form a composite 360-degree view of refuse vehicle **10**. One or more cameras can monitor a state of the hopper volume **132**, or may be coupled to various vehicle portions to avoid or reduce obstructed views. In some embodiments, cameras **510** make up some and/or all of sensors **610** that provide sensor data to a controller included in the ADAS. As shown in FIG. **14**, cameras **510** may be integrated into refuse vehicle **10** itself. For example, application kit **80** and/or cab **40** of refuse vehicle **10** may be modified such that integrated cameras **510** are installed into and protected by application kit **80** and/or cab **40** while still able to obtain appropriate image data. The cameras **510** may be disposed at any number of locations throughout and/or around the refuse vehicle **10**. While six cameras **510** are shown in FIG. **14**, it should be understood that the number and position of cameras **510** in 360° system **500** might vary without departing from the scope of the present invention. In some embodiments, cameras **510** include a front camera **512** and a rear camera **514** as part of 360° system **500**. In some embodiments, cameras **510** may be configured to detect an object and/or obstacle in the way of the vehicle **10**. The cameras may be in communication with a user interface in the cab **40**. The user interface may be configured to provide notifications regarding the object and/or obstacle in the way of the vehicle **10**.

[0069] Refuse vehicle **10** is shown on a vehicle axis system with an x-axis **1002** and y-axis **1004** that follow the International Organization for Standardization (ISO) Road Vehicles-Vehicle Dynamics and road-holding ability-Vocabulary (ISO Standard No. 8855:2011) Vehicle Axis System 2.10 convention, published December 2012, the entirety of which is herein incorporated by reference. The x-axis **1002** is a horizontal axis parallel to the vehicle's heading and in the forward direction of the vehicle such that it is also parallel to refuse vehicle **10**'s longitudinal plane of symmetry. The y-axis **1004** is perpendicular to the x-axis **1002** and the refuse vehicle **10**'s longitudinal plane of symmetry and is in the left direction of the vehicle of refuse vehicle **10**.

[0070] In some embodiments, the front camera **512** and the rear camera **514** are approximately positioned on the x-axis **1002** and at approximately the same height from the ground as the other cameras **510** such that the cameras all lie in the approximately same z-plane parallel to and above the xy-plane.

[0071] According to an exemplary embodiment shown in FIG. 15, ADAS includes a radar detection system, shown as radar system **600**, configured to detect the position, speed, direction of travel, and/or acceleration of one or more objects external to refuse vehicle **10**. Radar system **600** includes radar sensors, shown as radar sensors **610** integrated into the body of refuse vehicle **10**, with field of views, shown as radar FOVs **620**. Such integration can include mechanical or electrical coupling, wherein the electrical coupling can employ wired or wireless links. In some embodiments, radar sensors **610** make up some and/or all of sensors **610** that provide sensor data to controller. In some embodiments, radar sensors **610** are dual sensing radar sensors, and may have multiple radar FOVs **620**, such as a first FOV for short range sensing, shown in FIG. 15 as short range FOV **612**, and a second FOV for long range sensing, shown in FIG. 16 as long range FOV **614**. In some embodiments, the short range FOV **612** is wider than the long range FOV **614**. For example, the short range FOV **612** can be 45 degrees and the long range FOV **614** can be 20 degrees. Still in other embodiments the radar sensors **610** are single-distance radar sensors and have only a single FOV. In some embodiments, the radar sensors **610** are installed low to the ground (e.g., below an uppermost portion of the chassis **20** or a location of a camera **510**) in a substantially horizontal plane parallel to the xy-plane made by the x-axis **1002** and the y-axis **1004**. For example, the radar sensors **610** can be placed between 35 and 43 inches off the ground on a horizontal plane. Radar sensors **610** should be placed approximately horizontally to ensure they function properly. Placing radar sensors **610** low to the ground helps radar sensors **610** detect smaller vehicles such as motor cycles and smaller pedestrians. Placing sensors too low can cause collisions or interference (e.g., loss of line-of-sight) with the ground or road debris. Radar sensors **610** can provide sensor data to a controller of the ADAS indicating the existence of objects external to the refuse vehicle **10**. The objects can include pedestrians, vehicles, refuse containers, etc. In some embodiments, the sensor data includes a position, direction of travel, speed, and/or acceleration of detected objects and/or obstacles. The sensors **610** may be in communication with a user interface in the cab **40**. The user interface may be configured to provide notifications to a user via the user interface in the cab **40** regarding the object and/or obstacle in the way of the vehicle **10**.

[0072] The controller of the ADAS can detect one or more objects engaged with the refuse vehicle, such as via lift forks **146**, grabber fingers **166**, or the like. For example, the controller can store one or more predefined shapes, symbols, or other indicia of an object engaged with the refuse vehicle **10**. The controller can store an outline or dimension of a refuse bin or dumpster, recognize an object engaged with the refuse vehicle, and determine that the detected object is not an obstacle. Thus a detection thereof may be labeled or masked as a false event. In some embodiments, the controller can receive a position of an element of the refuse vehicle **10**, such as the position of the lift forks **146** or grabber fingers **166**. The controller can infer a location of an object based on the location thereof. For example, upon an actuation of a front lift fork **146**, the controller can mask all or a portion of a FOV of a front facing sensor (e.g., a camera FOV **520**, radar FOV **620**, or the like). A correlation between the objects (e.g., dumpsters) and the lift assembly may include a determination that the detect object is engaged with or otherwise related to the lift assembly.

[0073] According to an exemplary embodiment, radar system **600** includes two radar sensors **610** positioned on the front of cab **40** and positioned to be forward facing. In some embodiments, two radar sensors **610** are positioned on the front corners of cab **40** and positioned outward, with a bias towards facing towards the rear of refuse vehicle **10**. In some embodiments, two radar sensors **610** are positioned behind the cab **40** in a lower battery box area. Application kit **80** may also include on or more radar sensors **610**. In some embodiments, two radar sensors **610** integrated into the rear of application kit **80** and positioned to face outward are configured to detect obstacles behind the refuse vehicle **10**. Still in some embodiments, two radar sensors **610** can be integrated into the rear corners of application kit **80** and positioned to face an approximately 45 degree angle to the radar sensors **610** positioned to face behind the refuse vehicle **10**. While certain radar sensors **610** are shown in the configuration described above, it should be understood that the number and position

of radar sensors **610** in radar system **600** may vary without department from the scope of the invention. For example, radar system **600** may only include front-facing and rear-facing radar sensors **610**.

[0074] As shown in FIG. **16**, refuse vehicle **10** may include both 360° system **500** and radar system **600**. In some embodiments, a controller of the ADAS receives inputs from both 360° system **500** and radar system **600** and integrates the two inputs into a single composite data model of refuse vehicle **10** and/or its surroundings. Camera FOVs **520** and radar FOVs **620** may overlap. In some embodiments, the vehicle **10** includes a user interface in communication with the integrated 360° system **500** and the radar system **600**. In some embodiments, other sensors may be included in the application kit **80** such as LiDAR, Infrared, or the like. The user interface may be configured to display one or more notifications associated with potential collisions with objects and/or obstacles in the path of the vehicle **10**. In this way, the integrated system may be configured to provide notifications to a user interface in the cab **40** of the vehicle **10** in the event that a collision is about to occur based on the detection of the 360° system **500**, the radar system **600**, a combination of the two, or other sensors.

A. Notification Muting

[0075] Referring now to FIG. **17**, a process **1600** for muting (e.g., suppressing) or allowing notifications regarding sensor data is shown, according to an exemplary embodiment. In some embodiments, process **1600** is performed by one or more components of refuse vehicle **10**. For example, process **1600** can be performed by the controller of the ADAS.

[0076] In some embodiments, process **1600** includes providing a refuse vehicle including an ADAS system having one or more sensors and one or more controllable elements (step **1602**). Controllable elements may be the same or similar to the lift assembly **140**. Further, controllable elements may include the lift assembly **160**, the chute **238**, the outriggers **252**, the ladder assembly **254**, the monitor **264**, the monitor **314**, the boom assembly **354**, and/or the cab **404**. It should be appreciated that the controllable elements may further include a prime mover, steering components, power transmission or driver components, braking components, lift assemblies, electric actuators, hydraulic actuators, electric motors, systems, subsystems, assemblies, and/or any other components of refuse vehicle **10** controllable by an operator. In some embodiments, provided sensor can be the same or similar to sensors **610**. In some embodiments, the sensors can include 360° system **500**, radar system **600**, and collision detection system.

[0077] In some embodiments, process **1600** includes obtaining data from the one or more sensors and the one or more controllable elements relating to a detected event (step **1604**). In some embodiments, the data obtained includes sensor data and/or control data. The sensor data can include image data, proximity data, and or other types of data such as a direction of travel, speed, or acceleration of detected objects. The control data can include the position, direction of movement, speed, and/or acceleration of lift assembly **140** or other vehicle portions. Elements of the control data referring to a current or previous condition of the vehicle **10** can be referred to herein as a state of the vehicle **10**. The control data may include a list of past control signals provided to lift assembly **140**. In some embodiments, the data is obtained from the one or more sensors in network communication (e.g., via a wired or wireless network). For example, the network can include an Ethernet connection. The sensors may include cameras **510** and radar sensors **610**. The sensors can all be connected to the network for transmitting information to a controller. The network may include Ethernet connections employing copper or coax lines (e.g., differential twisted pairs). In some embodiments, the network employs fiber-optic lines. In some embodiments, the detected event is the presence of an obstacle. For example, the cameras **510** may detect the event or the controller may generate the event responsive to image data received by the cameras and conveyed to the controller. Various sensor data relating to the event may be provided to the controller. The controller may provide an indication of the event or information relevant to the event (e.g., a time, direction, speed, or the like) to a user of the lift device via a notification

generated by one or more control signals, as discussed in greater detail below in regard to step **1614**. In some instances, the detected event is the operation of the lift assembly **140** detected by the sensors (for example, the cameras **510**). It may be advantageous to mask a notification in such instances that would otherwise be provided to the user of the lift device that would otherwise be generated. For example, the user may already be aware of lift assembly movements they are controlling. The detected event may be correlated with the control data. For example, the control data can include the data received from one or more controllable elements at the time the sensor data indicated the presence of the obstacle. In some embodiments, the detected event may be based on a user input. For example, the detected event can be a movement of a joystick of the user interface referenced above with regard to FIGS. **14**, **15**, and **16**. In some embodiments, all sensor data is filtered by the corresponding control data. For example, the controller may compare sensor data to control data constantly, or according to a loop or interrupt, to identify where the sensor data includes false events based on the control data instances. Such false events may be correlated with a determination to mask the notification that would otherwise be provided to the user of the lift device. Thus, the false events can include instances of a detection of a portion of a vehicle, such as a lift assembly **140**, by a sensor of a vehicle, such that the ADAS provides an indication of an obstacle based on the detection of the portion of the vehicle.

[0078] In some embodiments, the lift assembly **140** or other vehicle portion may obscure a detection zone of a sensor (e.g., a field of view of a camera **510**) such that the sensor cannot monitor a portion of the vehicle surrounding the refuse vehicle **10**. According to some implementations, another sensor of a same or different type may monitor a relevant detection zone. For example, a front loading refuse vehicle **10** raising a refuse container can obscure a FOV of a front facing camera **512** of the refuse vehicle. Another sensor, such as a front facing emitter or sensor **610** of a radar system **610** may monitor the obscured zone. For example, the radar sensor **610** can be mounted lower on the refuse vehicle **10** such that the radar sensor **610**, in combination with the front facing camera **512**, can continuously monitor the detection zone or a portion thereof. In some implementations, the combination of vehicle sensors may not monitor the detection zone or a portion thereof. For example, the refuse container of the previous example can obscure one or more sensors such that the ADAS can determine that the sensors are blocked by a portion of the refuse vehicle **10**, but may not detect an obstacle beyond the refuse container. The controller can determine whether or how to mask a notification based on a determination that the ADAS may not detect an obstacle. For example, the ADAS can delay or fail to mask the notification, partially mask the notification, provide another indication of a non-operation of the ADAS, etc. In some embodiments the controller can determine that a detection zone is unobscured upon a failure of any sensor to detect their respective portion thereof (e.g., to preserve redundancy). In some embodiments, the controller can determine that a detection zone is unobscured upon a failure of every sensor, or a portion of sensors (e.g., to increase ADAS availability).

[0079] The controller can mask a notification based on an operational status of the refuse vehicle. An operational status can include a vehicle speed, gear selection, occupancy, brake status, or throttle of the refuse vehicle **10**. For example, the controller can mask a notification when the vehicle is stationary, when the brake is applied, or when the a gear selector is placed in park. Like other portions of the present disclosure, the operational status can be employed in combination with aspects provided herein. For example, the controller can determine whether or how to mask a notification based on the operational status in combination with an obscured sensor (e.g., where at least one sensor is obscured and the refuse vehicle is stationary, the controller may mask a notification).

[0080] In some embodiments, process **1600** includes filtering the sensor data through the control data (step **1606**) for the purpose of muting notifications. The controller may be configured to filter the sensor data through the vehicle control data to identify, remove, and/or tag false events from the sensor data that would otherwise be provided to a user of the vehicle **10** as a notification. False

notifications may be instances of sensor data that appear to indicate one or more objects are present around refuse vehicle **10**, but actually are due to refuse vehicle **10** itself and/or one or more of its components, or an element interfacing therewith (e.g., a refuse container engaged with lift forks **146** of the refuse vehicle **10**). The filter process can include using the control data to identify the position of one or more components of refuse vehicle **10** and comparing that position to the detected object in the sensor data. Sensor data observations that align with control data can be filtered out as false notifications. For example, the cab **40** may interface with the lift assembly **140** in front of refuse vehicle **10**. Forward radar can detect lift assembly **140** as an object and provide sensor data to controller indicating the position, direction of movement, speed, and/or acceleration of the object. Controller can also receive control data indicating that lift assembly **140** is interfaced with the refuse vehicle and the arms of cab **40** are lowered. If the sensor data is not filtered, controller may analyze the sensor data determine an object is present. If the sensor data is filtered through the control data, controller can compare the filter sensor data and the control data and determine the sensor data is a false event due to the lift assembly **140**, and that no external object is present. Accordingly, the controller can mask the notification that would otherwise be provided as a detection of an external object.

[0081] In some embodiments, process **1600** includes checking if the event is a false event (step **1608**). In some embodiments, if a false event is detected, the sensor data the notification is based on may be masked (e.g., removed, tagged, ignored, and/or adjusted by controller). For example, during the filtering process, controller can tag all sensor data that is determined to be due to one or more components of refuse vehicle **10** based on the control data as false event data, and mask (e.g., suppress or append) one or more control signals based on the false event data, therefore muting the notification that would otherwise occur. In some embodiments, if the event is determined to be a false event, the controller masks the notification and the process **1600** proceeds to step **1616** and ends. In other embodiments, if the event is determined to be a false event, the controller partially masks the notification and the process **1600** proceeds to step **1618** as described in greater detail below. In some embodiments, if the event is determined not to be a false event, process **1600** includes proceeding to step **1610**.

[0082] In some embodiments, process **1600** includes generating, via a user interface, a notification based on the sensor data (step **1610**). The notification may be a visual notification via a display (i.e., instrument display, console display) located within the cab **40**, and/or an auditory notification via notification a device in the cab **40**. The systems and methods herein can employ speakers, bells, air horns, relays, and other audio output devices used in vehicle systems to generate the audible signals. In some embodiments, the notification includes a recommended control action for a user to perform. For example, ADAS via radar system **600** including radar sensors **610** may detect a vehicle in a blind spot of refuse vehicle **10**. Controller can generate a notification to a driver indicating the presence of the vehicle. For another example, the refuse vehicle **10** is stopped, and ADAS senses fast approaching objects from the rear of refuse vehicle **10**. Controller can generate visual, audible, and/or haptic notifications that are apparent from outside of refuse vehicle **10** and/or the notifications themselves are external refuse vehicle **10** to notification those around refuse vehicle **10** of the approaching objects. In some embodiments, the notifications are audible natural language based notifications. Audible natural language based notifications can explain with language (according to a user preference for example) the content of the notification. For example, the notification may include an audible natural language based notification saying "Vehicle in blind spot." Natural language based notifications allow a user to understand what a notification is for without any other supplemental information. In some embodiments, the notifications can indicate information about refuse vehicle **10**. For example, notifications may include a tire pressure of refuse vehicle **10** while operating.

[0083] In some embodiments, process **1600** includes checking if the notification is cleared (step **1612**). For example, a user may clear a notification via a user input in user interface. Notifications

may also be cleared automatically by controller. In some embodiments, controller automatically clears notifications if the underlying event that triggered the notification is no longer detected. For example, a notification of a car in a blind spot of refuse vehicle **10** may persist so long as the car is in the blind spot. Once the car leaves the blind spot, controller may automatically clear the notification. In some embodiments, if the notification is cleared, process **1600** proceeds to step **1616** and ends. In some embodiments, if the notification is not cleared, process **1600** proceeds to step **1614**.

[0084] In some embodiments, process **1600** includes generating one or more control signals based on the sensor data and control data. Controller can be configured to generate control signals based on the sensor data and control data in response to a true event (i.e., not a false event). Control signals may be commands to operate refuse vehicle **10** and/or one or more components of refuse vehicle **10**. For example, sensor data indicate a vehicle ahead of refuse vehicle **10**, and control data may indicate refuse vehicle **10** is traveling at a sufficient speed that it will collide with the vehicle if the speed is not diminished. Controller can provide a clearable notification a driver of the vehicle. If the clearable notification is not cleared (i.e., before a time threshold, wherein the time threshold is the point in time determined by controller where action must be taken to avoid a collision), controller can generate control signals to activate the brakes of refuse vehicle **10** and prevent the collision. In some embodiments, the control signals may also control components of refuse vehicle **10** including actuators, motors, lift assemblies, etc. In some embodiments, process **1600** skips steps **1610** and **1612**, as described above in regards to muting notifications, and proceeds directly to generating one or more control signals via at **1618**. For example, the controller may be configured to provide a visual warning instead of both an audio and visual notification on the user interface in the event that the event is determined to be a false event, rather than muting the notification entirely and proceeding directly to step **1616**. In this sense, a notification that is determined to be a false event may be partially masked to be less obtrusive to a user of the vehicle **10**, while also providing awareness via the visual aspect of the notification. In other embodiments, Controller can be configured to automatically generate one or more control signals in emergencies where there is not enough time to generate a notification and wait for the clearable notification to be cleared. For example, controller may determine that a control action such as emergency braking should be taken immediately in order to avoid an accident. In some embodiments, after generating the control signals process **1600** proceeds to step **1616** and ends.

[0085] As utilized herein, the terms “approximately,” “about,” “substantially”, and similar terms are intended to have a broad meaning in harmony with the common and accepted usage by those of ordinary skill in the art to which the subject matter of this disclosure pertains. It should be understood by those of skill in the art who review this disclosure that these terms are intended to allow a description of certain features described and claimed without restricting the scope of these features to the precise numerical ranges provided. Accordingly, these terms should be interpreted as indicating that insubstantial or inconsequential modifications or alterations of the subject matter described and claimed are considered to be within the scope of the disclosure as recited in the appended claims.

[0086] It should be noted that the term “exemplary” and variations thereof, as used herein to describe various embodiments, are intended to indicate that such embodiments are possible examples, representations, or illustrations of possible embodiments (and such terms are not intended to connote that such embodiments are necessarily extraordinary or superlative examples).

[0087] The term “coupled” and variations thereof, as used herein, means the joining of two members directly or indirectly to one another. Such joining may be stationary (e.g., permanent or fixed) or moveable (e.g., removable or releasable). Such joining may be achieved with the two members coupled directly to each other, with the two members coupled to each other using a separate intervening member and any additional intermediate members coupled with one another, or with the two members coupled to each other using an intervening member that is integrally

formed as a single unitary body with one of the two members. If “coupled” or variations thereof are modified by an additional term (e.g., directly coupled), the generic definition of “coupled” provided above is modified by the plain language meaning of the additional term (e.g., “directly coupled” means the joining of two members without any separate intervening member), resulting in a narrower definition than the generic definition of “coupled” provided above. Such coupling may be mechanical, electrical, or fluidic.

[0088] References herein to the positions of elements (e.g., “top,” “bottom,” “above,” “below”) are merely used to describe the orientation of various elements in the FIGURES. It should be noted that the orientation of various elements may differ according to other exemplary embodiments, and that such variations are intended to be encompassed by the present disclosure.

[0089] The hardware and data processing components used to implement the various processes, operations, illustrative logics, logical blocks, modules and circuits described in connection with the embodiments disclosed herein may be implemented or performed with a general purpose single- or multi-chip processor, a digital signal processor (DSP), an application specific integrated circuit (ASIC), a field programmable gate array (FPGA), or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A general purpose processor may be a microprocessor, or, any conventional processor, controller, microcontroller, or state machine. A processor also may be implemented as a combination of computing devices, such as a combination of a DSP and a microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration. In some embodiments, particular processes and methods may be performed by circuitry that is specific to a given function. The memory (e.g., memory, memory unit, storage device) may include one or more devices (e.g., RAM, ROM, Flash memory, hard disk storage) for storing data and/or computer code for completing or facilitating the various processes, layers and modules described in the present disclosure. The memory may be or include volatile memory or non-volatile memory, and may include database components, object code components, script components, or any other type of information structure for supporting the various activities and information structures described in the present disclosure. According to an exemplary embodiment, the memory is communicably connected to the processor via a processing circuit and includes computer code for executing (e.g., by the processing circuit or the processor) the one or more processes described herein.

[0090] The present disclosure contemplates methods, systems and program products on any machine-readable media for accomplishing various operations. The embodiments of the present disclosure may be implemented using existing computer processors, or by a special purpose computer processor for an appropriate system, incorporated for this or another purpose, or by a hardwired system. Embodiments within the scope of the present disclosure include program products comprising machine-readable media for carrying or having machine-executable instructions or data structures stored thereon. Such machine-readable media can be any available media that can be accessed by a general purpose or special purpose computer or other machine with a processor. By way of example, such machine-readable media can comprise RAM, ROM, EPROM, EEPROM, or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other medium which can be used to carry or store desired program code in the form of machine-executable instructions or data structures and which can be accessed by a general purpose or special purpose computer or other machine with a processor. Combinations of the above are also included within the scope of machine-readable media. Machine-executable instructions include, for example, instructions and data which cause a general purpose computer, special purpose computer, or special purpose processing machines to perform a certain function or group of functions.

[0091] Although the figures and description may illustrate a specific order of method steps, the order of such steps may differ from what is depicted and described, unless specified differently

above. Also, two or more steps may be performed concurrently or with partial concurrence, unless specified differently above. Such variation may depend, for example, on the software and hardware systems chosen and on designer choice. All such variations are within the scope of the disclosure. Likewise, software implementations of the described methods could be accomplished with standard programming techniques with rule-based logic and other logic to accomplish the various connection steps, processing steps, comparison steps, and decision steps.

[0092] It is important to note that the construction and arrangement of the vehicle **10** and the systems and components thereof as shown in the various exemplary embodiments is illustrative only. Additionally, any element disclosed in one embodiment may be incorporated or utilized with any other embodiment disclosed herein. Although only one example of an element from one embodiment that can be incorporated or utilized in another embodiment has been described above, it should be appreciated that other elements of the various embodiments may be incorporated or utilized with any of the other embodiments disclosed herein.

Claims

1. A system comprising: a vehicle comprising: a chassis; a body assembly coupled to the chassis; a prime mover configured to generate mechanical energy to drive the vehicle; and a lift assembly; and a vehicle control system comprising a sensor integrated into the body assembly, the vehicle control system is configured to: receive sensor data from the sensor, the sensor data indicating a potential event; generate a notification associated with the potential event; receive control data indicating a state of the vehicle; filter the sensor data through the control data to determine if the potential event is a false event associated with the state of the vehicle or a true event associated with an external source; initiate a control action in response to the potential event, wherein the control action provides the notification to a user of the vehicle responsive to a determination that the potential event is the true event; and mask the notification in response to a determination that the potential event is the false event, wherein to mask the notification, the vehicle control system is configured to: mask an audible output of the notification; and maintain a visual output of the notification.
2. The system of claim 1, wherein the sensor data comprises an indication of a portion of the vehicle undergoing displacement relative to the chassis and an object external to the vehicle, and wherein the vehicle control system is configured to: determine a correlation between the object external to the vehicle with the portion of the vehicle undergoing displacement; and mask the notification responsive to the determination of the correlation.
3. The system of claim 1, wherein the sensor includes: a radar system comprising a radar sensor; and a camera system comprising at least one forward facing camera disposed vertically above the radar sensor.
4. The system of claim 3, wherein each sensor is in network communication with a user interface comprising an audio output device disposed in a cab of the vehicle.
5. The system of claim 1, wherein the vehicle control system is configured to predict the potential event based on a position, direction of travel, and speed of a detected object.
6. The system of claim 1, wherein the control data comprises an indication of a movement or a position of the lift assembly.
7. A method, comprising: receiving, by a vehicle control system, sensor data indicating a potential event from a sensor, wherein the sensor is integrated into a body assembly coupled to a chassis of a vehicle comprising: a prime mover configured to generate mechanical energy to drive the vehicle; and a lift assembly; generating, by the vehicle control system, a notification associated with the potential event; receiving, by the vehicle control system, control data indicating a state of the vehicle; filtering, by the vehicle control system, the sensor data through the control data to determine if the potential event is a false event associated with the state of the vehicle or a true

- event associated with an external source; and masking, by the vehicle control system, the notification in response to a determination that the potential event is the false event by: masking an audible output of the notification; and maintaining a visual output of the notification.
- 8.** The method of claim 7, further comprising: determining a correlation between an object external to the vehicle with a portion of the vehicle undergoing displacement; masking the notification responsive to the determination of the correlation, wherein the sensor data comprises an indication of the portion of the vehicle undergoing displacement relative to the chassis; and the sensor data comprises an indication of the object external to the vehicle.
- 9.** The method of claim 7, wherein the sensor includes: a radar system comprising a radar sensor; and a camera system comprising at least one forward facing camera disposed vertically above the radar sensor.
- 10.** The method of claim 9, wherein each sensor is in network communication with a user interface comprising an audio output device disposed in a cab of the vehicle.
- 11.** The method of claim 7, wherein the potential event is predicted based on a position, direction of travel, and speed of a detected object.
- 12.** The method of claim 7, wherein the control data comprises an indication of a movement or a position of the lift assembly.
- 13.** A non-transitory computer-readable media, comprising instructions stored thereon that, when executed by one or more processors, cause the one or more processors to: receive sensor data indicating a potential event from a sensor; generate a notification associated with the potential event; receive control data indicating a state of a vehicle; filter the sensor data through the control data to determine if the potential event is a false event associated with the state of the vehicle or a true event associated with an external source; initiate a control action in response to the potential event, wherein the control action provides the notification to a user of the vehicle responsive to a determination that the potential event is the true event; and mask the notification in response to a determination that the potential event is the false event, wherein, to mask the notification, the instructions cause the one or more processors to: mask an audible output of the notification; and maintain a visual output of the notification.
- 14.** The computer-readable media of claim 13, wherein the instructions include instructions to: determine a correlation between an object external to the vehicle with a portion of the vehicle undergoing displacement; and mask the notification responsive to the determination of the correlation, wherein: the sensor data comprises an indication of the portion of the vehicle undergoing displacement relative to a chassis of the vehicle; and the sensor data comprises an indication of the object external to the vehicle.
- 15.** The computer-readable media of claim 13, wherein the sensor includes: a radar system comprising a radar sensor; and a camera system comprising at least one forward facing camera disposed vertically above the radar sensor.
- 16.** The computer-readable media of claim 13, wherein the instructions cause the one or more processors to predict the potential event based on a position, direction of travel, and speed of a detected object.
- 17.** The computer-readable media of claim 13, wherein the control data comprises an indication of a movement or a position of a lift assembly of the vehicle.
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